# Keeping the Boomers

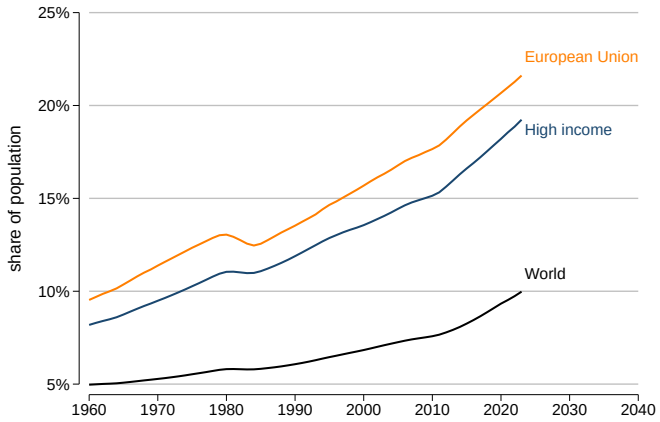
## Older Workers, Their Firms, and Their Coworkers after the Abolition of Early Retirement

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# Motivation: aging society




Share of the population aged 65 and over, World Bank (2024). In the EU it is above 20%.

Government response: increase the retirement age

# Firms have to keep their older workers longer

**He may help**

- Complementary skills in production (Kremer, 1993)
- Coworkers learn from experienced colleagues (Jarosch et al., 2021)
- Losing him is costly when his skills are specific (Jäger and Heining, 2024)

**He may hurt**

- Holds a place in the promotion queue (Bianchi et al., 2023)
- Firms hire fewer young workers (Martins et al., 2009)
- Deferred pay: earns more than he produces (Lazear, 1979)

**Question: What happens to the firms and coworkers if older workers stay longer?**

- Hard to measure, because the retirement decision is not random

I use a natural experiment from Hungary to answer this question.

- Abolition of early retirement from 1 January 2012, and gradual increase in the general retirement age
  - Men born 1950 and 1951: retire at 60, in 2010–2011
  - Men born 1953: retire at 63, in 2016
  - Exogenous variation in retirement time by birth cohort
- Compare firms with the same number of men born 1950, 1951 or 1953 but different numbers born in 1953
- Two levels of outcomes
  - Firms: workers, output, and productivity
  - Coworkers: retention and wages

For every older worker kept at the firm

**Firms**

- Number of workers increases by 13%
- Hires and exits do not differ (except the older men's own exits)
- Value added increases by 16%, value added per worker rises by 7%
  - This is driven by low-skill older worker retention

**Coworkers**

- Retention: 2.1 pp decrease, but not significant; too imprecise to tell
- Wage
  - same occupation: +5.3%
  - low-skill coworkers: +3.5%
  - men: +3.6%

- **Worker exit**
  - Exits are costly: hiring and training, firm-specific skills *(Abowd and Kramarz, 2003; Bartel et al., 2014; Becker, 1962; Muehlemann and Pfeifer, 2016)*
  - External hires are not perfect substitutes *(Jäger et al., 2024; Jaravel et al., 2018; Yanadori and Kato, 2007)*
  - Spillover effects on incumbent workers *(Azoulay et al., 2010; Waldinger, 2012)*
  - **Contribution:** exit of older workers
- **Productivity of older workers**
  - Diminishing productivity *(Cardoso et al., 2011; Lovász and Rigó, 2013)*
  - High wages relative to productivity *(Lazear, 1979; Frimmel et al., 2018)*
  - **Contribution:** effect on firm-level productivity
- **Pension reforms**
  - Firms: fewer young hires, lower output *(Martins et al., 2009)*; more young workers, same productivity *(Carta et al., 2021)*; middle-aged displaced *(Boeri et al., 2022)*
  - Coworkers: older workers block promotions and hires *(Bianchi et al., 2023; Badalyan, 2025)*
  - **Contribution:** firm and coworker outcomes at the same time

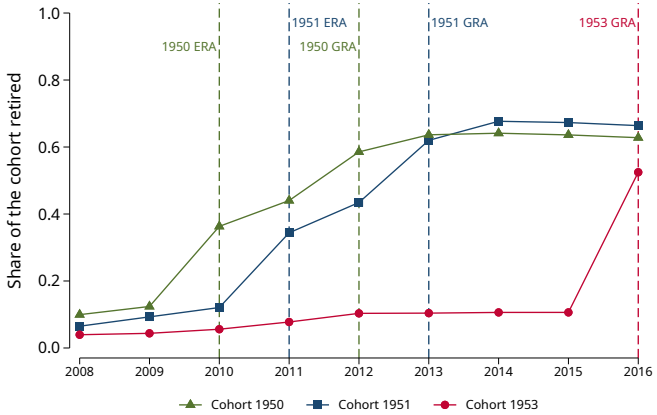
- Mandatory pay-as-you-go pension system: active workers finance the benefits of retired people
- Eligibility: 20 years of service; benefit depends on past earnings
- Until 2012: two statutory retirement ages
  - Early retirement age (ERA): 60, reduced benefit (full benefit with 40+ years)
  - General retirement age (GRA): 62, full benefit
- From the 1952 cohort, the GRA rises by half a year per cohort, to 65 for those born 1957; first applied in 2014, when the 1952 cohort reached 62.5
- Most men retire at the early age *(Bíró and Elek, 2018)*
- From 2011, Women40: women can retire after 40 years of service, with full benefit *(Simonovits and Tir, 2017)*
- Retiring means leaving the labor market because of the earnings cap
- Costly to fire older workers: dismissal protection within 5 years of the GRA

# The abolition split men born two or three years apart

| Born | Early age (year) | General age (year) | Could leave from |
|---|---|---|---|
| 1950 | 60 (2010) | 62 (2012) | 2010 |
| 1951 | 60 (2011) | 62 (2013) | 2011 |
| 1953 | none | 63 (2016) | 2016 |

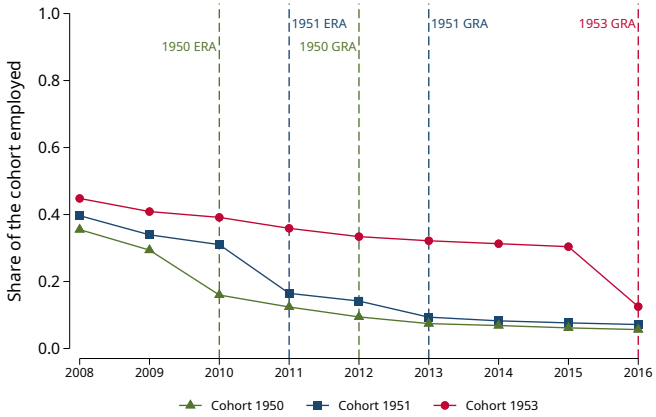
- Announced November 2011; anyone 60 by the end of 2011 could still claim
- Abolished early retirement for both genders, but women can still use Women40
- 1952 transitional cohort
  - Turned 60 in 2012, after the abolition
  - Could leave at 60 if a termination agreement was signed before 2012, or for arduous work
  - Share retired rose from 11% in 2011 to 23% in 2012: partly kept, partly gone
  - I follow Bárdits et al. (2026) and exclude this cohort

# 1953 men cannot retire for 3 more years




Share of each cohort retired (received first pension transfer) in December.

# Retiring means leaving work




Share of each cohort employed in December.

- ADMIN3: administrative monthly panel of 50% of the population, 2003–2017
- Yearly balance sheets, every firm that uses double-entry bookkeeping in Hungary
- I aggregate to a yearly panel, taking the Decembers
- **Focal men:** men born 1950, 1951 or 1953 at the firm in December 2009

# Sample selection

| | |
|---|---|
| *Firms* | |
| Employed a man born in 1950, 1951, or 1953 in December 2009 | 16,514 |
| Every such man also there in December 2007 and 2008 | 10,191 |
| 5 to 50 workers in the 50% sample in December 2009 | 5,327 |
| At least one year of balance sheet data | 4,218 |
| *Workers* | |
| Focal men in the estimation sample | 6,033 |
| Coworkers at the sample firms in December 2009 | 63,734 |

# Comparison: firms with the same number of focal men, different numbers born in 1953

| | Focal men born in 1953 | | | | |
|---|---|---|---|---|---|
| Focal men | 0 | 1 | 2 | 3+ | All |
| 1 | 1,763 | 1,256 | | | 3,019 |
| 2 | 279 | 426 | 127 | | 832 |
| 3+ | 62 | 146 | 115 | 44 | 367 |
| All | 2,104 | 1,828 | 242 | 44 | 4,218 |

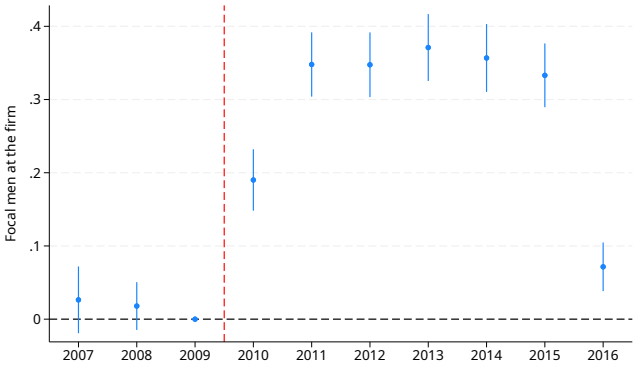
# Firms and focal men look the same in 2009; coworkers nearly so

| | Mean, comparison | Coefficient | Standard error | Observations |
|---|---|---|---|---|
| *A. Firms* | | | | |
| Number of workers | 15.4 | −0.018 | (0.289) | 4,218 |
| Part-time share | 0.1 | −0.002 | (0.006) | 4,218 |
| ln sales | 12.7 | 0.023 | (0.032) | 4,133 |
| ln value added | 11.3 | 0.017 | (0.030) | 3,998 |
| ln value added per worker | 8.1 | 0.005 | (0.022) | 3,972 |
| *B. Coworkers* | | | | |
| Age | 41.6 | −0.261** | (0.120) | 63,734 |
| Female | 0.3 | −0.011* | (0.006) | 63,734 |
| ln wage | 11.8 | 0.006 | (0.014) | 62,616 |
| High-skill job | 0.3 | 0.002 | (0.006) | 63,565 |
| White-collar job | 0.4 | 0.002 | (0.006) | 63,565 |
| *C. Focal men* | | | | |
| ln wage | 11.9 | −0.002 | (0.016) | 5,884 |
| High-skill job | 0.3 | −0.017 | (0.011) | 6,014 |
| White-collar job | 0.3 | −0.015 | (0.011) | 6,014 |
| Part-time | 0.1 | −0.011 | (0.008) | 6,014 |

$$F_{jt} = \beta \cdot T_j \times Post_t + \gamma_j + \delta_{kt} + \mu_{nt} + \varepsilon_{jt} \qquad \text{first stage}$$

$$y_{jt} = \theta \cdot F_{jt} + \gamma_j + \delta_{kt} + \mu_{nt} + e_{jt} \qquad \text{second stage}$$

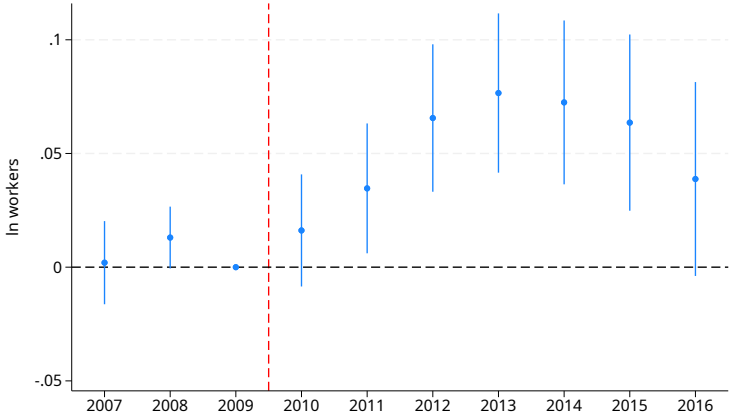
- $y$: ln number of workers, ln value added, ln value added per worker
- $T_j$: number of focal men born in 1953 at firm $j$ in December 2009
- $Post_t = 1$ in 2010–2015
- $F_{jt}$: focal men at the firm in year $t$
- Fixed effects: firm, industry $\times$ year, focal men in 2009 $\times$ year
- IV: $F_{jt}$ instrumented by $T_j \times Post_t \Rightarrow$ effect of keeping one more man




Focal men at the firm per additional 1953 man, relative to 2009.

| Pooled 2010–2015 | |
|---|---|
| 1953 men × post | 0.310*** |
| | (0.023) |
| First-stage $F$ | 312 |

# Firms with a 1953 man employ more workers from 2011 on




ln workers in the 50% sample per additional 1953 man, relative to 2009.

# The firm "grows" mainly because its older men do not leave

| Per additional 1953 man, total 2010–2015 | Hires | Exits |
|---|---|---|
| Under 30 | 0.49 | 0.42 |
| | (0.48) | (0.40) |
| 30–49 | 1.08 | 0.84 |
| | (0.70) | (0.56) |
| 50–59 | 0.39 | 0.15 |
| | (0.28) | (0.22) |
| 60 and over | −0.03 | −0.46*** |
| | (0.08) | (0.08) |

# The firm keeps one more focal man, and output rises

| Per man kept, 2010–2015 | OLS | Reduced form | IV | Observations |
|---|---|---|---|---|
| ln workers | 0.254*** | 0.048*** | 0.130*** | 34,368 |
| | (0.010) | (0.014) | (0.036) | |
| ln value added | 0.170*** | 0.057*** | 0.159*** | 32,415 |
| | (0.011) | (0.018) | (0.050) | |
| ln value added per worker | 0.000 | 0.027** | 0.074** | 32,133 |
| | (0.007) | (0.013) | (0.036) | |

Note: ln workers is the December count in ADMIN3; value added per worker divides by the balance-sheet number of workers (yearly average).

# Heterogeneity by the skill of the focal worker

| IV, per man kept | High-skill | Low-skill | Observations |
|---|---|---|---|
| ln workers | 0.185** | 0.117*** | 34,292 |
| | (0.091) | (0.041) | |
| ln value added | 0.077 | 0.190*** | 32,341 |
| | (0.129) | (0.055) | |
| ln value added per worker | −0.034 | 0.102*** | 32,059 |
| | (0.090) | (0.039) | |
| First stage | 0.244*** | 0.336*** | |
| | (0.023) | (0.016) | |

High-skill: FEOR 1–3; low-skill: FEOR 4–9, in December 2009.

$$F_{jt} = \beta \cdot T_j \times Post_t + \alpha_i + \delta_{kt} + \mu_{nt} + \varepsilon_{it} \qquad \text{first stage}$$

$$y_{it} = \theta \cdot F_{jt} + \alpha_i + \delta_{kt} + \mu_{nt} + e_{it} \qquad \text{second stage}$$

- $y$: still at the 2009 firm, ln wage
- $T_j$: number of focal men born in 1953 at his firm $j$ in December 2009
- $Post_t = 1$ in 2010–2015
- $F_{jt}$: focal men at his 2009 firm in year $t$
- Fixed effects: worker, industry $\times$ year, focal men in 2009 $\times$ year, of his 2009 firm
- IV: $F_{jt}$ instrumented by $T_j \times Post_t \Rightarrow$ effect of keeping one more man

# Coworkers' wages do not fall; retention too imprecise to tell

| Per man kept, 2010–2015 | OLS | Reduced form | IV | Observations |
|---|---|---|---|---|
| Still at the 2009 firm | $0.074^{***}$ | $-0.006$ | $-0.021$ | 573,606 |
| | (0.007) | (0.007) | (0.026) | |
| ln wage | $0.017^{***}$ | $0.008^{*}$ | $0.028^{*}$ | 478,134 |
| | (0.002) | (0.005) | (0.016) | |
| ln wage, incumbents | $0.011^{***}$ | 0.004 | 0.011 | 371,274 |
| | (0.002) | (0.005) | (0.013) | |

| IV, per man kept | Still at the firm | ln wage | ln wage, incumbents |
|---|---|---|---|
| *By the skill of the coworker* | | | |
| Low-skill | −0.018 | 0.035** | 0.017 |
| | (0.027) | (0.017) | (0.014) |
| High-skill | −0.027 | 0.010 | −0.005 |
| | (0.035) | (0.025) | (0.020) |
| *By occupation* | | | |
| Same occupation | −0.032 | 0.053** | 0.033* |
| | (0.042) | (0.026) | (0.020) |
| Other occupations | −0.014 | 0.020 | 0.003 |
| | (0.026) | (0.017) | (0.014) |

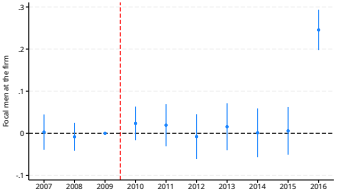
| IV, per man kept | Still at the firm | ln wage | ln wage, incumbents |
|---|---|---|---|
| *By the coworker's age* | | | |
| Under 30 | −0.011 | 0.057* | 0.021 |
| | (0.033) | (0.031) | (0.024) |
| 30 to 49 | −0.027 | 0.021 | 0.010 |
| | (0.029) | (0.018) | (0.014) |
| 50 to 59 | −0.039 | 0.017 | 0.007 |
| | (0.034) | (0.020) | (0.016) |
| 60 and over | 0.078 | 0.075 | 0.037 |
| | (0.064) | (0.095) | (0.077) |
| *By gender* | | | |
| Men | −0.013 | 0.036** | 0.016 |
| | (0.028) | (0.018) | (0.014) |
| Women | −0.045 | −0.003 | −0.003 |
| | (0.036) | (0.024) | (0.019) |

- Keeping one more older man at the firm raises its number of workers by 13%, mainly because the older men do not leave
- Value added rises by 16% and value added per worker by 7%, driven by low-skilled men
- Coworkers' wages rise in the man's own occupation (+5.3%), in low-skill jobs (+3.5%) and among men (+3.6%)
- Robustness checks
  - **Placebo:** 1953 against 1955 men
  - **Endogenous variable:** focal men still at the firm

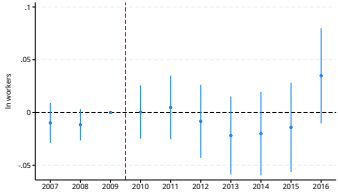
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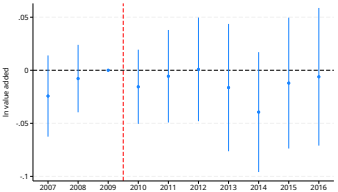
# Backup: placebo, 1953 against 1955 men: two years of age alone move nothing



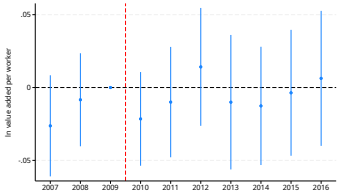

First stage: focal men at the firm




ln workers




ln value added




ln value added per worker

back

# Backup: placebo against main design, pooled 2010–2015

| Reduced form, per man | Placebo (1955 vs 1953) | Main (1953 vs 1950–51) |
|---|---|---|
| First stage | 0.011 | 0.310*** |
| | (0.027) | (0.023) |
| ln workers | −0.002 | 0.048*** |
| | (0.015) | (0.014) |
| ln value added | −0.001 | 0.057*** |
| | (0.020) | (0.018) |
| ln value added per worker | 0.004 | 0.027** |
| | (0.014) | (0.013) |

back

# Backup: December 2009 focal men still at the firm as the endogenous variable

| IV, per man kept | Focal men at the firm (main) | December 2009 focal men still there |
|---|---|---|
| *Firms* | | |
| First stage | $0.310^{***}$ | $0.305^{***}$ |
| | (0.023) | (0.017) |
| ln workers | $0.130^{***}$ | $0.134^{***}$ |
| | (0.036) | (0.038) |
| ln value added | $0.159^{***}$ | $0.162^{***}$ |
| | (0.050) | (0.051) |
| ln value added per worker | $0.074^{**}$ | $0.076^{**}$ |
| | (0.036) | (0.036) |
| *Coworkers* | | |
| First stage | $0.290^{***}$ | $0.304^{***}$ |
| | (0.033) | (0.023) |
| Still at the 2009 firm | $-0.021$ | $-0.020$ |
| | (0.026) | (0.025) |
| ln wage | $0.028^{*}$ | $0.027^{*}$ |
| | (0.016) | (0.015) |
| ln wage, incumbents | 0.011 | 0.011 |
| | (0.013) | (0.013) |

back